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Passive force balancing of an active magnetic regenerative liquefier

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ABSTRACT

Active magnetic regenerators (AMR) have the potential for high efficiency cryogen liquefaction. One active magnetic regenerative liquefier (AMRL) configuration consists of dual magnetocaloric regenerators that reciprocate in a persistent-mode superconducting solenoid. Issues with this configuration are the spatial and temporal magnetization gradients that induce large magnetic forces and winding currents. To solve the coupled problem, we present a force minimization approach using passive magnetic material to balance a dual-regenerator AMR. A magnetostatic model is developed and simulated force waveforms are compared with experimental measurements. A genetic algorithm identifies force-minimizing passive structures with virtually ideal balancing characteristics. Implementation details are investigated which affirm the potential of the proposed methodology.

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1. Introduction

Although hydrogen (H^2) has a gravimetric energy density several times greater than common fossil fuels such as gasoline or diesel, the low volumetric energy density of gaseous hydrogen has motivated research efforts on liquid hydrogen [1] and liquefaction technologies [2]. The low process efficiency of state-of-the-art liquefiers coupled with the added expense of long-distance transport in cryogenic tankers significantly raise the cost of liquid hydrogen at distributed refueling stations.

The active magnetic regenerator (AMR) uses a magnetocaloric material (MCM) as the matrix media in a thermal regenerator [3], and shows promise for high efficiency distributed cryogen liquefaction [4]. In an AMR, each differential regenerator section undergoes an independent Brayton refrigeration cycle consisting of: (1) adiabatic magnetization, (2) isofield heat rejection, (3) adiabatic demagnetization and (4) isofield heat absorption. In operation, warm fluid is pumped to the hot end at T_H where heat is released and cold fluid is pumped to the cold end at T_C where heat is absorbed.

Room temperature AMR devices using permanent magnets have demonstrated commercially relevant cooling powers [5], efficiencies [6] and temperature spans [7]; however, costs must be reduced for market penetration [8,9]. While room temperature AMR devices are an active area of research, active magnetic regenerative liquefiers (AMRL's) are less mature and only a small number of cryogenic devices using superconducting magnets (SC) have been reported. Zimm et al. (1996) [10] measured a 35 K temperature span while rejecting heat to liquid nitrogen (LN2). Rowe and Tura (2006) [11] measured a 50 K temperature span from room temperature using a three material regenerator, and the device was later modified for cryogenic testing [12]. The layered experiments were investigated in subsequent analytic and numerical works [13–15]. Kim et al. (2013) [16] presented a no-load cold temperature of 24 K while rejecting heat to LN2 using 83 grams of magnetocaloric material. The AMRL was recently retrofitted with a GdBCO high temperature superconducting solenoid [17]. The device was numerically investigated and an optimized layering composition was proposed [18]. While the temperature spans reported by Kim et al. [16] approached the domain of a hydrogen liquefier, larger devices are required to provide commercially relevant capacities.

Barclay et al. (2016) [19] and Holladay et al. (2017) [20] described a large-scale AMRL with an ultimate goal of hydrogen liquefaction from room temperature. While a temperature span of 100 K was reported with 2.1 kg of a single magnetocaloric material, the varying permeability in the solenoid core induced an electromotive force across the copper superconducting stabilizer. This ultimately heated the superconducting winding towards

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Nomenclature

Roman

A	area [m^2]
B	magnetic flux density [T]
F	force [N]
H	magnetic field strength [A m^{-1}]
I	solenoid winding current [A]
j	current density [A m^{-2}]
M	magnetization [A m^{-1}]
$\dot{Q}$	heat transfer [W]
r	radius [m]
t	time [s]
$\underleftrightarrow{T}$	Maxwell electromagnetic stress tensor [N m^{-2}]
T	temperature [K]
V	volume [m^3]
z	spatial coordinate along solenoidal axis

Greek

δ_{ij}	Kronecker delta [–]
Γ	geometric factor [–]
μ_0	Permeability of free space [H/m]
ρ	density [kg m^{-3}]
ρ_e	electrical resistivity [Ωm]
σ	specific magnetization [A m^2 kg^{-1}]

Subscripts and superscripts

C	cold reservoir or cold side
coil	superconducting magnet winding
Curie	magnetic ordering temperature
Eddy	Eddy current
H	hot reservoir or hot side
m	middle passive structure
magnet	position of AMR relative to magnet
o	outer passive structure

the critical temperature, limiting the applied field strength to 3.3 T. The magnet heating was found to increase with regenerator mass, applied field strength and operating frequency.

Improved cooling capacities require increased regenerator mass which, to avoid a magnet quench, decreases the magnetic field strength and consequentially the cooling capacity; a challenge with reciprocating AMRL configurations. The magnetic field from the coupled regenerator-solenoid system must be explored to reduce spatial and temporal magnetization gradients. Barclay et al. (1986) [21] simulated force waveforms for reciprocating and rotary configurations. Rowe and Barclay (2002) [22] used the centerline field of a static, air-bored solenoid simulation to evaluate magnetization and magnetic forces. An optimization routine found a flywheel configuration minimizing the cycle RMS torque accounting for magnetic, pumping and inertial loads. While forces were balanced at the drive input, a flywheel does not resolve the magnetization gradients and coupled magnet heating problem.

Peksoy and Rowe (2005) [23] later performed magnetostatic field simulations to investigate the variation of magnetization in a single and two-material AMR. Rowe and Tura (2008) [24] continued this work by investigating ferromagnetic shims to concentrate magnetic field lines in the regenerator, demonstrating that the influence of magnetic material on the magnetic field distribution can be both the detrimental and beneficial. Recently, Mira et al. (2017) [25] solved the magnetostatics problem to investigate demagnetization in gadolinium (Gd) regenerators.

Arnold et al. (2011) [26] reported experimental measurements of the mechanical, eddy and magnetic work in a reciprocating AMR device. Although large forces were present, it was found that the thermodynamic cycle work was on the order of the experimental uncertainty. This emphasized that while *regenerator* efficiencies can be high, *device* efficiencies may be heavily penalized without force balancing.

While Peksoy and Rowe (2005) [23] and Mira et al. (2017) [25] solved the magnetostatics problem, several works have investigated magnetic forces with a simplified treatment of the magnetic field distribution (i.e. $\vec{B} = \mu_0 \vec{H}$). Kamiya et al. (2006) [27] analyzed the force waveform of a reciprocating AMR with two gadolinium doped dysprosium aluminum garnet regenerators using a similar methodology as Rowe and Barclay (2002) [22]. The authors reported a 60 % force reduction using magnetic material between regenerators. Allab et al. (2006) [28] simulated the magnetic force on a Gd sample as a function of the local magnetic field strength, and presented force waveforms for a magnetized and demagnetized regenerator. Balli et al. (2011) [29] and Gama et al. (2016) [30] compared experiments and simulations of the force on magnetic material using a similar formulation.

In the present work, a 2-D, axisymmetric magnetostatic model is developed to study the interaction of a multilayered AMR and superconducting magnetic field generator. Magnetic forces are analyzed and compared with experiments on the AMR device described by Barclay et al. (2016) [19] and Holladay et al. (2017) [20]. A passive, soft ferromagnetic structure is proposed to balance both spatial and temporal magnetization gradients in an effort to simultaneously address the force balancing and magnet heating problems impeding AMRL development. An optimization is formulated to find passive geometries minimizing the force waveform of a dual regenerator assembly. The optimized force waveform is discussed and implementation details such as solution sensitivity and field distribution are investigated.

2. Methodology

2.1. AMR configuration

Passive force balancing is investigated on the AMR configuration shown in Fig. 1 [19,20]. A persistent-mode, conduction-cooled NbTi Cryomagnetics 70–650-010CF superconducting solenoid is used to generate the magnetic field for two regenerators mounted axially opposite to each other onto a common cold heat exchanger (CHEX). The solenoid consists of two composite windings with the properties listed in Table 1.

Each regenerator contains eight layers of magnetocaloric material which are summarized in Table 2 and shown in Fig. 3. The layers consist of rare-earth gadolinium and gadolinium alloys with yttrium, terbium, erbium, dysprosium and holmium with compositions selected for a respective Curie temperature spacing of 20 K per layer. Spherical particles of each refrigerant are prepared by AMES using a rotating disk apparatus [20] and packed into monolithic regenerators with an approximate porosity of 0.36 and mean particle size of 225 μm.

A linear actuator drives the regenerator assembly with constant velocity, as shown in Fig. 1, and a second actuator drives a double-acting piston displacing 2520 cm_3 of pressurized Helium. The two

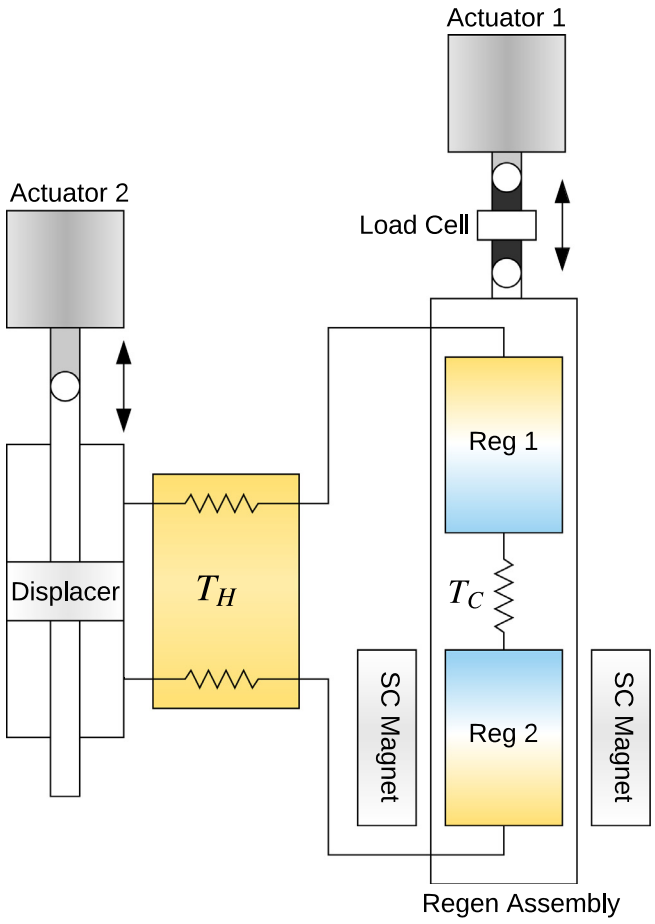


Fig. 1. Schematic of experimental AMR device. Actuator 1 drives regenerator assembly inside stationary superconducting (SC) magnet while load cell measures net force. Actuator 2 displaces pressurized helium through a temperature controlled circulator at T_H.

actuators are driven out of phase with equal displacement times of 2 s, yielding a trapezoidal waveform with a cycle frequency of 0.125 Hz. A Futek LCM350 load cell measures the net assembly force with a calculated uncertainty of 160 N. In practice, however, the sensor is able to measure the static assembly weight within 10 N. The measured force waveforms are corrected for the assembly weight of 0.33 kN. E-type thermocouples measure the temperature across each layer with an approximate uncertainty of 0.5 K. Measurements are recorded with a National Instruments CompactDAQ.

2.2. Magnetostatics model

The superconducting winding is modeled in COMSOL Multiphysics using the AC/DC magnetic fields interface in a 2-D axisymmetric domain. As the AMRL cycle period generally exceeds 1 s, Maxwell's equations reduce to:

$$\nabla \times \vec{H} = \vec{j} \tag{1}$$

$$\nabla \cdot \vec{B} = 0 \tag{2}$$

where $\vec{B}$ is the magnetic flux density in T, $\vec{H}$ is the magnetic field strength in A/m and $\vec{j}$ is the current density in the winding cross section in A/m^2. These are solved numerically with the constitutive relation $\vec{B} = \mu_0(\vec{H} + \vec{M})$, where M is the magnetization in A/m. Boundary conditions are imposed to satisfy magnetic flux conservation ($\vec{n} \cdot \vec{B} = 0$).

The simulations described here consider gadolinium-like materials with molecular mean field theory (MFT) [31,32] generated magnetization data shifted to the respective Curie temperature. The specific magnetization, defined as the ratio of magnetization (M) to density (ρ), is shown as a function of temperature and field (μ_0H) in Fig. 2. The magnetization is then corrected for porosity and processed into relative permeability curves. Fixed heat rejection and absorption temperatures of T_H = 280 K and T_C = 120 K are considered, relevant for the first stage of a hydrogen liquefier or natural gas liquefaction. As passive force balancing cannot reduce the thermodynamic cycle work stemming from regenerative temperature perturbations, a linear, field-independent axial temperature profile is assumed in each regenerator.

Magnetic forces are evaluated in COMSOL Multiphysics by numerically integrating the Maxwell electromagnetic stress tensor:

$$\vec{F} = \oint_S \overleftrightarrow{T} \cdot \vec{n}\, ds \tag{3}$$

where S is a surface in free space enclosing the considered body and $\vec{n}$ is a unit vector normal to the integration surface. The Maxwell stress tensor, $\overleftrightarrow{T}$, is defined for any coordinate system as:

$$\overleftrightarrow{T}_{ij} = \frac{B_i B_j}{\mu_0} - \delta_{ij} \frac{||\vec{B}||_2^2}{2\mu_0} \tag{4}$$

where δ_{ij} is the Kronecker delta and $||\vec{B}||_2^2$ is the squared norm of the magnetic flux density. Bjørk et al. (2010) [33] and Meessen et al. (2013) [34] used this formulation to evaluate forces in permanent magnet assemblies using analytic field expressions derived from Fourier series solutions of the governing partial differential equations.

If the flux density and magnetization are uniform over a material volume, and the applied field and magnetization vectors reduce to components along the solenoidal axis (z), the force can be described as:

$$F_z = \mu_0 M_z \frac{\partial H_z}{\partial z} \tag{5}$$

which is referred to the Kelvin force [35,31]. Although the more-robust Maxwell stress tensor formulation is used in the present work, the Kelvin force familiarizes the physical mechanisms describing magnetic forces in magnetized bodies. Eq. (5) describes a force increase with field gradient and magnetization, which occur with increased coil current and decreased magnetocaloric material temperature, respectively.

2.3. Optimization formulation

Passive, soft ferromagnetic structures are proposed in the regenerator assembly to minimize variations in magnetization. These passive structures fill the void regions on either side of the regenerators in Fig. 1, and an optimization routine determines the structure geometry ($r_{m1}, r_{m2}, r_{o1}, r_{o2}$) that balances magnetic forces. Fig. 3 shows the parameterized outer radii, and only the

Table 1
Superconducting magnet dimensions.

Winding	Inner diameter [mm]	Outer diameter [mm]	Length [mm]	Turns [–]
1	172.7	180.6	203.2	1708
2	180.6	232.9	203.2	17791

Table 2
Rare-earth alloys synthesized by AMES laboratory for eight-layer regenerator.

Layer	Composition	T_{Curie} [K]	Mass [g]	Diameter [mm]	Length [mm]
L1	$Gd_{0.16}Ho_{0.84}$	153	57	31.8	14.2
L2	$Gd_{0.27}Ho_{0.73}$	173	100	38.1	17.3
L3	$Gd_{0.15}Dy_{0.85}$	193	139	44.5	17.7
L4	$Gd_{0.32}Dy_{0.68}$	213	172	50.8	16.8
L5	$Gd_{0.67}Er_{0.33}$	232	202	57.1	15.6
L6	$Gd_{0.3}Tb_{0.7}$	253	235	63.5	14.7
L7	$Gd_{0.91}Y_{0.09}$	274	258	69.9	13.3
L8	Gd	293	268	76.2	11.6

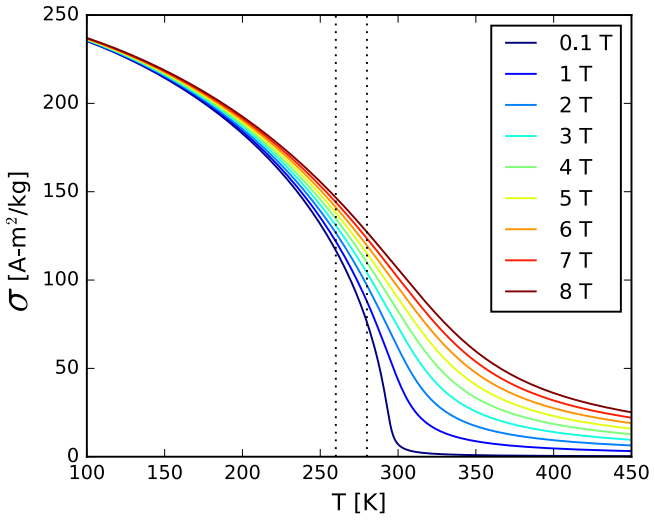


Fig. 2. Specific magnetization of gadolinium (σ), generated with MFT, illustrating a second-order, para-ferromagnetic phase change at T_{Curie}=293 K. Magnetization increases with field and decreases with temperature. σ is shown for fields (μ_0H) ranging from 0.1 to 8 T. Vertical dashed lines indicate layer operating range.

top half of the assembly is shown. The middle passive structure (r_{m1}, r_{m2}) has a fixed length of 127 mm with a 10 mm gap on either side for the layer 1 (L1) housing and flow distributor. The outer structure (r_{o1}, r_{o2}) has a fixed length of 85 mm, constrained by the superconducting magnet's bore, and is spaced 10 mm from layer 8 (L8). A fixed inner radius of 10 mm is considered in both structures. The passive structure is composed of 1010 steel with experimental B-H data provided by COMSOL Multiphysics.

The force waveform, $F(z_{magnet})$, is composed of static force simulations as the regenerator assembly translates downwards with increasing values of z_{magnet}. An optimization problem is formulated to find the structure geometry defined by ($r_{m1}, r_{m2}, r_{o1}, r_{o2}$) that minimizes the l^2 norm of the force waveform:

$$\text{Min}\|F(z_{magnet}, r_{m1}, r_{m2}, r_{o1}, r_{o2})\| = \sqrt{\sum_{z_{magnet}=0}^{z_{max}} F(z_{magnet}, r_{m1}, r_{m2}, r_{o1}, r_{o2})^2} \quad (6)$$

Due to the problems' non-convexity, the optimization is performed in MATLAB using a genetic algorithm. An initial population of 25 randomly generated configurations are evaluated, and the 4 design candidates with the highest fitness (lowest l^2 norm) are cloned into the next generation. Remaining design candidates are procreated with a crossover fraction of 0.9 in a process mimicking

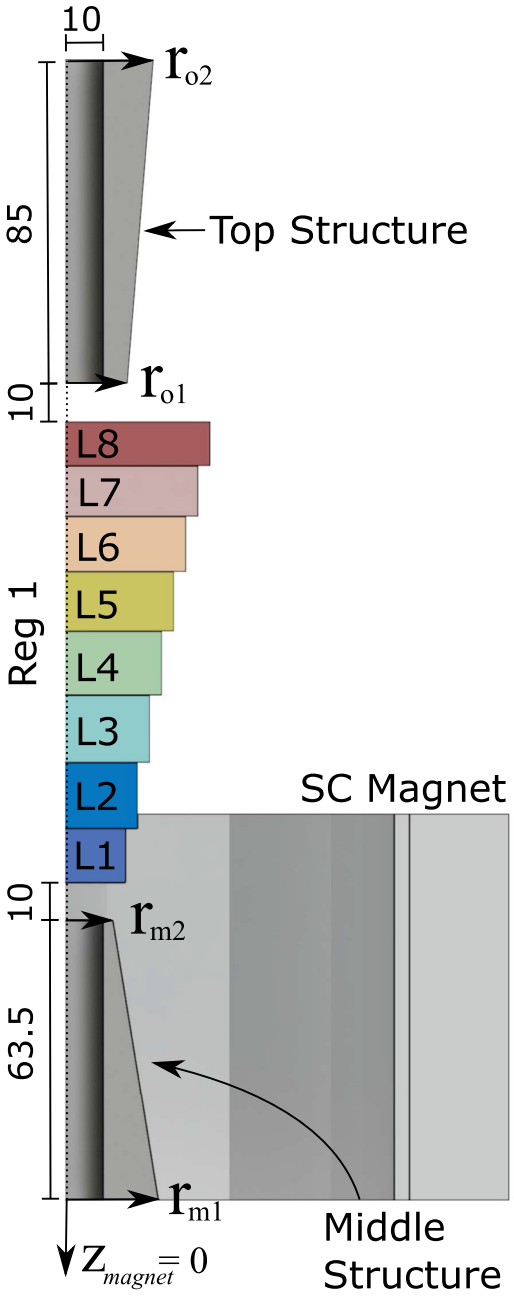


Fig. 3. AMR configuration (Reg 1 with 8 layers), composite superconducting magnet and parameterized passive structures. Fixed dimensions have units of mm. Figure shows axisymmetric domain, and due to symmetry only the top half of the assembly is shown.

biological evolution, amalgamating the design traits of two favorable parents. The probability of a design procreating into the next generation increases with fitness, and designs with high forces are eliminated. Further implementation details are outlined in the MATLAB documentation. With appropriate bounds on the design vector, the optimization converges within 12 h using an AMD FX-8350 CPU.

3. Results

3.1. Validation

Two experiments are performed to validate the force waveform; one with regenerators in the paramagnetic (PM) phase at $\mu_0 H = 3$ T ($I_{coil} = 33.4$ A) and a second in the ferromagnetic (FM) phase at $\mu_0 H = 4$ T ($I_{coil} = 44.2$ A). $\mu_0 H$ refers to the magnetic flux density in the absence of magnetic material. For the ferromagnetic test, the assembly is cooled with saturated nitrogen vapor at atmospheric pressure (LN2 boil-off). Fig. 4 shows the average temperatures of each layer over one cycle for the top (red) and bottom (blue) regenerators in the paramagnetic (dashed) and ferromagnetic (solid) phases. The Curie temperature of each layer is shown for reference.

Fig. 5 shows the measured and simulated force waveform using the experimentally measured temperatures of each magnetic material as shown in Fig. 4. The force waveform for the paramagnetic (PM) and ferromagnetic (FM) phases are shown in Fig. 5 (A) and (B), respectively. The solid black line shows the simulated waveform, while the red and blue dashed lines show experimental measurements with the linear actuator moving in the positive (→) and negative(←) directions. After the regenerator is displaced in either direction, a double-acting piston drives heat transfer fluid through the material matrix as shown in Fig. 1. The perturbed solid temperature invokes a magnetization change in each layer of magnetic refrigerant, causing the offset between red and blue curves. This manifests as the required thermodynamic cycle work and serves as a limit to what can be attenuated with passive force balancing.

The model results are in good agreement with the magnitude of the measured force waveforms, and the peak force location is predicted for both experiments. While the simulated paramagnetic forces are nearly identical to measurements, the ferromagnetic simulations are 12.5 % higher than measured. The discrepancy is likely caused by a combination of simulated magnetization data

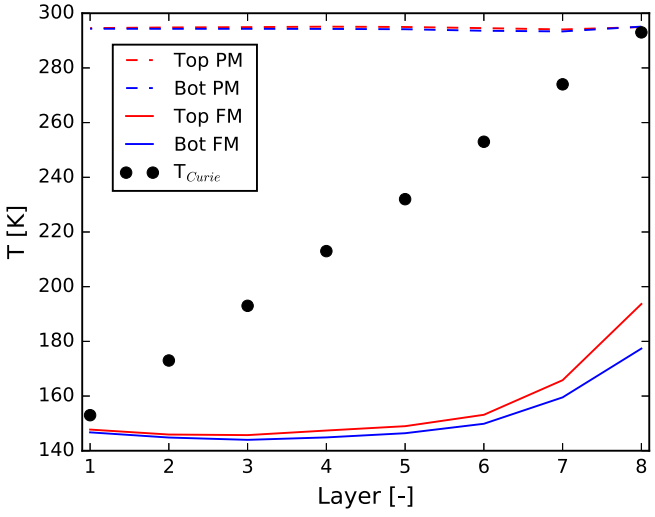


Fig. 4. Measured temperatures for validation in paramagnetic (PM) and ferromagnetic (FM) phases. (For interpretation of the references to colour in this figure caption, the reader is referred to the web version of this article.)

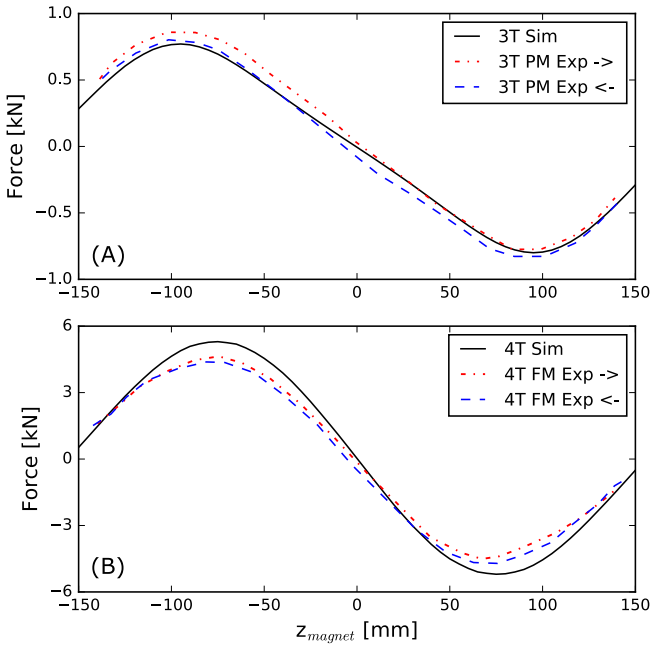


Fig. 5. Validation of simulated waveforms (solid black) with measurements for paramagnetic (A) and ferromagnetic (B) conditions. Red and blue curves correspond to positive and negative linear actuator velocities, where the difference yields the thermodynamic cycle work. (For interpretation of the references to colour in this figure caption, the reader is referred to the web version of this article.)

and the magnetic alloy treatment. The MFT data in Fig. 2 shows higher ferromagnetic magnetization at low field strengths than experimental measurements [36]. Additionally, shifting gadolinium magnetization data to the ordering temperatures of layers 1–8 causes properties to be referenced 140 K from the Curie temperature, where the impact of varying alloy parameters (e.g. electron spin and orbital angular momentum) on magnetization is most pronounced [31]. This error is expected to decrease in operation with a linear temperature span from $T_H = 280$ K to $T_C = 120$ K, where materials operate in the vicinity of the magnetic ordering temperature.

3.2. Force minimization

The optimization converged on a design vector of $(r_{m1}, r_{m2}, r_{o1}, r_{o2}) = (25.45,\ 12.66,\ 16.41,\ 23.25)$ mm, shown in Fig. 3, for a winding current of 66.0 A ($\mu_0 H = 6$ T) and temperature reservoirs of 280 K and 120 K. Fig. 6 shows the magnetic force waveform of the top regenerator (solid red), bottom regenerator (solid blue), top passive structure (dashed red), middle passive structure (dashed black) and bottom passive structure (dashed blue). Forces on the middle passive structure are out of phase with forces on the top and bottom regenerators. As such, the middle passive structure is responsible for the majority of force cancellation while outer structures (red and blue dashed lines) fine-tune the net waveform shape. Although the large magnetic forces exerted on individual components requires detailed housing design, the superposition or net assembly force (solid black) is effectively reduced to zero throughout the AMR cycle.

4. Discussion

4.1. Solution sensitivity

The passive structures are optimized for $T_H = 280$ K, $T_C = 120$ K and $\mu_0 H = 6$ T; however, the impact of operating conditions on

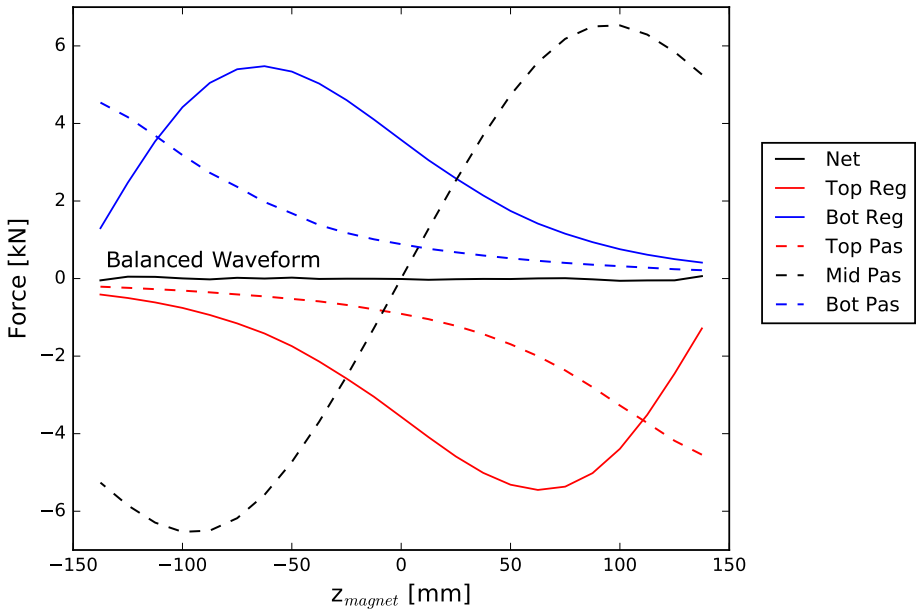


Fig. 6. Force contribution from each component at μ_0H = 6 T in optimized geometry. *Reg* denotes regenerator and *Pas* denotes passive magnetic material. Superposition of forces demonstrates near-ideal cancellation (Balanced Waveform) at this operating condition. (For interpretation of the references to colour in this figure caption, the reader is referred to the web version of this article.)

the force waveform with passive structures must be investigated. Fig. 7 shows contours of the maximum net force over the waveform as a function of μ_0H and cold side temperature for the optimized assembly, while Fig. 8 shows contours of the maximum net force with two regenerators. The contours illustrate the force variation during the transient cooling process at any field strength.

Without passive balancing material, the maximum net force increases weakly with decreasing temperature and strongly with increasing field strength as shown in Fig. 8. Fig. 7 shows the insensitivity of net force to field strength at the design temperature of 120 K; the same passive structure balances forces at an applied field of 3 T (33.2 A) and 6 T (66 A). Furthermore, the peak net force is significantly reduced for nearly all operating conditions with passive balancing material.

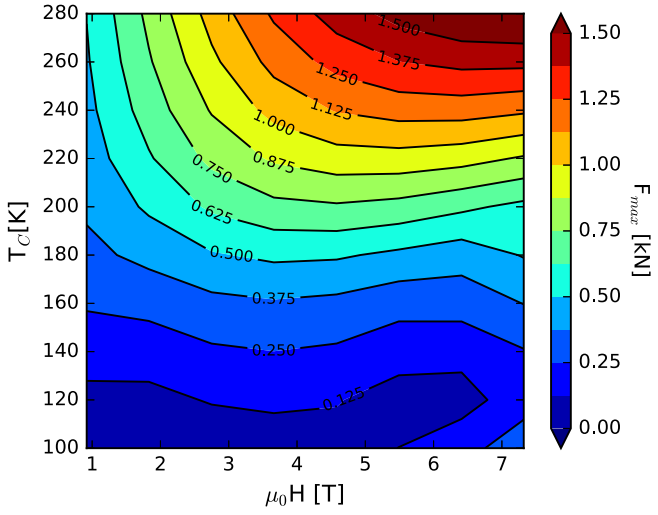


Fig. 7. Contours of the simulated maximum net force as a function of the cold side temperature and flux density. Passive structures are optimized for T_C = 120 K and μ_0H = 6 T. Hot side temperature fixed at 280 K.

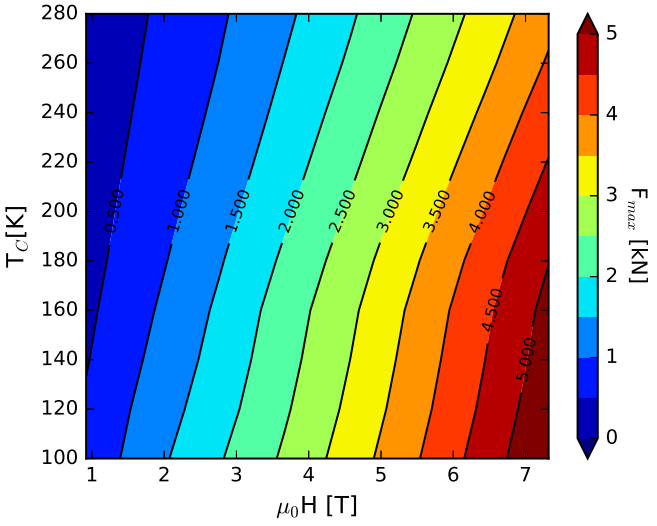


Fig. 8. Contours of the simulated maximum net force as a function of the cold side temperature and flux density for dual regenerator system (no passive balancing).

4.2. Field homogeneity

It is important to minimize the field strength during the demagnetized blow, as the adiabatic temperature change scales with $B^{2/3}$ for gadolinium alloys [37]. Fig. 9 shows the centerline field distribution for μ_0H = 6 T with an air bore (black), with magnetocaloric material (red) and with the optimized assembly (blue). The regenerator locations are indicated by vertical dashed lines.

The magnetocaloric material compresses field lines in the solenoid bore, increasing the high field strength. This phenomena was observed and exploited by Rowe and Tura (2008) [24], however Fig. 9 demonstrates how the same mechanisms can also be detrimental by increasing the flux density during the low-field fluid blow. Fortunately, the increased low field strength is minimal with the addition of passive balancing material.

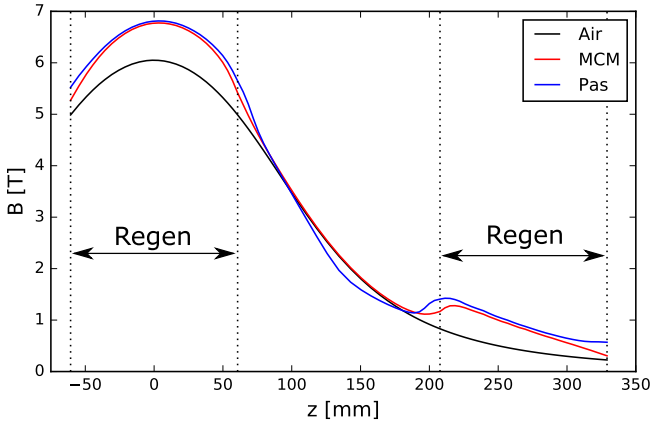


Fig. 9. Centerline magnetic flux density for air, regenerators (MCM) and the optimized assembly (Pas) showing a minimal increase in low field strength with passive balancing material. (For interpretation of the references to colour in this figure caption, the reader is referred to the web version of this article.)

4.3. Eddy currents

A consequence of the electrically conducting force balancing structure is the generation of eddy currents. The power dissipation from induced currents in an electrically resistive medium act as a parasitic load into the cold side of the AMR. Kittel (1990) [38] derived analytic expressions for eddy current power dissipation in primitive geometries

$$\dot{Q}_{Eddy} = \frac{\Gamma AV}{32\rho_e}(dB/dt)^2 \quad (7)$$

where Γ is a geometric form factor, A is the area enclosed by the largest possible current loop, V is the volume of material and ρ_e is the electrical resistivity. As AMR performance increases with field strength and operating frequency, $\Gamma AV/\rho_e$ must be minimized. While ΓAV can be reduced with thin laminations of the 1010 alloy considered here, advanced soft magnetic materials with high electrical resistivity deserve further investigation.

5. Conclusion

A passive, soft ferromagnetic structure is proposed to balance both spatial and temporal magnetization gradients in an effort to simultaneously address the force balancing and magnet heating problems found in large-scale, reciprocating AMRL devices. A magnetostatic model is developed and validated with experimental force measurements. A genetic algorithm is implemented to identify force-minimizing passive balancing structures with a cold side temperature and field strength of 120 K and $\mu_0 H = 6$ T, respectively.

The methodology produces a unique structure that decreases the net assembly force by a factor of 100. The optimized structure is shown to be effective over a wide range of operating conditions and has minimal impact on the effective magnetic field change. Future works will focus on the experimental treatment of magnetization properties, experimental testing of optimized passive structures and the minimization of low field strength.

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